

3(a)(i)	1 <u>promoter</u> 2 <u>operator</u> 3 <i>lac Z</i> / β-galactosidase gene 4 <i>lac Y</i> / lactose permease gene ; ;	2
3(a)(ii)	<i>β-galactosidase</i> – any one of: digests / hydrolyses, lactose to, monomers / glucose + galactose ; converts / isomerises, lactose to allolactose ; <i>lactose permease</i> increases / allows, lactose uptake / lactose entry / permeability (of cell) to lactose ;	2
3(b)(i)	any one from: (protein) produced all the time ; (protein) concentration does not vary (in response to molecular signals) ;	1
3(b)(ii)	any three from: 1 repressor (protein) not bound to, lactose / allolactose ; 2 repressor binds to <u>operator</u> ; 3 <u>RNA polymerase</u> cannot move to, operator / (structural) genes ; 4 no / prevents, transcription (of genes) / formation of mRNA ;	3
3(b)(iii)	any one from: proteins / enzymes, made, all the time / when not needed / too much / in excess / in uncontrolled fashion ; waste of, amino acids / ATP / nucleotides ; decrease growth ;	1